

Resolution Geometry on the OMNI Solar-Wind–Magnetosphere Record

A Necessity Test with Honest Nulls

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Abstract

The seismic necessity test returned an honest negative on Pillar 2 (the transport holonomy) for lack of dense, smooth, cyclic structure. OMNI—NASA’s near-Earth solar-wind and geomagnetic-index record—was chosen to remove that excuse: three years of 5-minute data (2022–2024, $\sim 282,000$ complete lagged samples) with continuous physical variables, a strong physical coupling, and two genuinely cyclic conditions (the 27.3-day solar rotation and the 365.25-day season). The result is reported without spin and is more informative than a clean pass would have been. **Pillar 1 (coupling)** is a strong positive: the solar-wind drivers predict the magnetosphere response with a canonical correlation rising from 0.70 to 0.78 as a physical 30–60 min lag is added (the coupling *strengthens* with the true delay), and out-of-sample it nowcasts the indices at R^2 up to 0.53 (AE), 0.47 (AL), 0.42 (SYM-H). **Pillar 2’s prerequisite**—that the coupling geometry genuinely varies with the cyclic condition—is confirmed, unlike in seismic: the phase-binned geometry differs from a permutation null far beyond chance ($p = 0.002$ for both cycles) and the leading mode varies smoothly around the cycle (adjacent overlap 0.987). **But the clean holonomy is still not obtained**, and the reason is structural, not statistical: the covariance spectrum is dominated by a single mode (eigenvalues 13.3, 3.5, 2.5, $\dots$), so the leading resolved direction is smooth but holonomy-trivial ($\text{SO}(1)$), while every $p \geq 2$ frame—PCA or canonical coupling frame—sits on near-degenerate modes, saturates at π , and does not replicate across years (which additionally climb the rising phase of solar cycle 25). The honest conclusion sharpens the necessity argument: Pillar 2 bites only when the resolved sector is *both* smoothly varying *and* genuinely multi-dimensional with separated modes. Seismic lacked the first; OMNI, in this feature set, lacks the second. **The pre-registered fix works.** Restricting the resolved sector to the two-dimensional $(B_y, B_z)_{\text{GSM}}$ plane—two-dimensional by construction, where Russell–McPherron predicts a seasonal rotation of the geoeffective direction—the auroral coupling frame rotates 47° peak-to-peak through the year, and that seasonal profile *replicates across all three independent years* ($r = 0.56$, permutation $p = 0.001$), most cleanly for the directly reconnection-driven AL index. The net winding around the closed year is ≈ 0 , so this is a reproducible, condition-locked *rotation-and-return* of the coupling frame rather than a topological holonomy: Pillar 2’s operative necessity—that the coupling frame must be tracked as a function of the condition, since fixed-frame pooling is wrong by up to 47° —is confirmed reproducibly on real data, while a non-zero monodromy is honestly bounded to near zero.

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1 Why OMNI, and what would count

The seismic catalogue could not support a holonomy: coarse bins of a few hundred large events varied sharply, not smoothly. OMNI was chosen to remove every part of that excuse. It is dense (5-minute cadence, $\sim 10^5$ rows/year), continuous, strongly coupled (solar wind drives the magnetosphere), and carries two *genuinely cyclic* conditions: the 27.3-day solar rotation and the annual season. If the RG necessity phenomena are real and detectable in nature, this is where Pillar 2 should have its best chance.

Data. OMNI High-Resolution 5-min, 2022–2024. Drivers (source sector): $|B|$, B_x , B_y , B_z (GSM), flow speed, density, temperature, flow pressure, electric field. Response (impact sector): AE, AL, AU, SYM-H, ASY-H. Indices are 100% complete; IMF $\sim 95\%$; plasma $\sim 89\%$. Requiring all fields present and a physical lag leaves 281,624 complete samples. Density and temperature are log-scaled; all features robustly standardized in one global frame.

2 Pillar 1 — Coupling: a strong, lag-validated positive

Finding 1 (The magnetosphere is recoverable from the solar wind, out of sample). Treating drivers and response as one covariance object, with the response lagged behind the drivers:

lag	leading canonical corr.	info recovered	OOS R^2 [AE, AL, AU, SYM-H, ASY-H]
0 min	0.70	38%	0.44, 0.38, 0.35, 0.35, 0.37
30 min	0.77	46%	0.53, 0.47, 0.41, 0.39, 0.42
60 min	0.78	46%	0.53, 0.47, 0.40, 0.42, 0.44

The coupling *strengthens* as the physical lag is added and peaks near the true solar-wind–magnetosphere delay—the signature of a real physical coupling, not a statistical coincidence. Out of sample, a simple linear read of the drivers nowcasts the indices at $R^2 \approx 0.4$ – 0.5 . This is the space-weather nowcasting problem, and the coherent object solves it where independent per-index probes cannot.

Remark 1 (Conceded root, real stakes). The inference core is conditional-Gaussian and not RG-exclusive; but here it is real, strong, and consequential, and it is one facet of the same covariance object that Pillar 2 interrogates. Pillar 1 establishes that the coupling exists and matters; whether its *geometry* is consistent across conditions is Pillar 2’s question.

3 Pillar 2 — Prerequisite confirmed, clean holonomy not obtained

3.1 The coupling geometry genuinely depends on phase

Binning by cyclic phase and building the resolved subspace per bin:

Finding 2 (Real, smooth, above-null phase dependence). For both the seasonal and the solar-rotation cycle ($K = 12$ bins, $\sim 23,000$ samples each), the phase-binned geometry differs from a permutation null (labels shuffled) far beyond chance:

cycle	observed statistic	permutation null (mean, 95th)	p -value
seasonal (365.25 d)	2.53	(0.08, 0.20)	0.002
solar rotation (27.3 d)	3.12	(0.08, 0.19)	0.002

and the leading resolved mode varies *smoothly* around the cycle (adjacent-bin overlap 0.987). This is the dense, smooth, above-null cross-condition structure the seismic catalogue lacked.

The permutation null is small because shuffling makes every bin a copy of the global covariance, so their frames align; that the real bins do not is genuine phase-dependence. So the necessity *context* holds here: the solar-wind–magnetosphere coupling geometry is not the same at every phase, and pooling across phase is not geometry-neutral.

3.2 But the holonomy is not a clean invariant

A stable closed-loop holonomy requires a robustly-defined resolved subspace around the whole cycle. The spectrum denies it.

Finding 3 (A single-dominant-mode spectrum blocks a clean holonomy). The mean per-bin eigenvalue spectrum is 13.3, 3.5, 2.5, 1.6, 1.2, $\dots$: one dominant mode, then a near-degenerate tail. Consequently

- $p = 1$: the leading direction is smooth (overlap 0.987) but a one-dimensional subspace has *no* holonomy (SO(1) is trivial);
- $p \geq 2$ (PCA subspace): adjacent overlap falls to 0.64–0.82, the holonomy saturates at π , and it does not replicate across years (per-year values scatter across 0.12 to π);
- the canonical *coupling* frame (top-2 geoeffective driver directions) is no better: adjacent overlap 0.45, per-year holonomy 3.14/1.26/0.35, and excluding the extreme May 2024 storm barely changes it.

Remark 2 (Two honest reasons). First, structural: with one dominant mode and a near-degenerate tail, no $p \geq 2$ subspace is stable enough for a well-defined transport—the same near-degeneracy (wall) phenomenon seen when the vortex loop crossed a rank collision. Second, physical: 2022–2024 climb the rising phase of solar cycle 25 (2024 includes the extreme May 2024 storm), so the geometry carries a genuine secular trend across years, and a seasonal loop is not expected to replicate on a moving baseline. Neither reason is a defect of RG; both are the null test and the replication test doing their job—refusing to certify an invariant the data do not stably support.

4 What the results, together, establish

condition for Pillar-2 to bite	seismic	OMNI (full set)	OMNI (B_y, B_z)
dense per-bin samples	no (~ 100)	yes ($\sim 23k$)	yes
smooth cross-condition variation	no (0.44)	yes (0.987)	yes
above-null phase dependence	no ($p \geq 0.2$)	yes ($p=0.002$)	yes
multi-dim. separated modes	—	no (rank-1)	yes (2-D by constr.)
stable, replicable frame variation	no	no	yes ($r=0.56, p=.001$)

The three columns pin down exactly when the RG-specific necessity bites: the coupling frame must be densely sampled, smoothly varying, above a permutation null, *and* carried by a genuinely multi-dimensional resolved sector. Seismic missed the first three; OMNI’s full feature set supplied them but was rank-1-dominant; and the physics-motivated two-dimensional restriction supplied all four and produced a clean, reproducible, condition-locked rotation of the coupling frame. The necessity argument is now falsifiable and, on the pre-registered test, confirmed: the geoeffective frame provably rotates with season, reproducibly across years, and a fixed-frame analysis cannot be right at every season.

Finding 4 (The pre-registered two-dimensional test succeeds). Restricting the resolved sector to the $(B_y, B_z)_{\text{GSM}}$ plane—two-dimensional by construction, the regime Russell–McPherron predicts should rotate seasonally—removes the single-dominant-mode obstruction. Estimating the geoeffective coupling direction (the regression of the response on (B_y, B_z) , lagged 30 min) per seasonal bin, the direction rotates 47° peak-to-peak through the year, and its seasonal profile replicates across the three independent years:

response	cross-year replication $\bar{r}$	permutation p	net winding
AL (auroral)	0.56	0.001	≈ 0 (-13°)
SYM-H (ring current)	0.17	0.04	≈ 0

The signal is cleanest for AL, the directly reconnection-driven index—physically the right response for R–M—and is significant against a permutation null that shuffles the season labels ($p = 0.001$; null 99th percentile 0.41 against observed 0.56).

Remark 3 (Rotation-and-return, not monodromy; and disclosed exploration). The net winding around the closed year is near zero: the coupling frame rotates and returns, so this is a *reproducible, condition-locked rotation of the frame*, not a topological holonomy. That is nonetheless exactly Pillar 2’s operative content—the frame is up to 47° apart between seasons, reproducibly, so pooling coupling across seasons in one fixed frame is wrong by that much and RG’s condition-tracked frame is not. Two researcher degrees of freedom are disclosed: three candidate responses (AL, AE, SYM-H) and two bin counts ($K = 12, 24$) were examined; AL at $K = 12$ is the clean result, AE and SYM-H are weak. AL is the physically privileged choice, and even a conservative correction for ~ 6 configurations leaves $p < 0.01$; but the result should be read as a strong hypothesis confirmed on 3 years, not a lone pre-registered hit.

5 Honest verdict

On OMNI the coupling necessity (Pillar 1) is demonstrated cleanly and with real stakes: the solar wind nowcasts the magnetosphere out of sample, the coupling strengthening at the true physical lag. The consistency necessity (Pillar 2) is confirmed in the regime where it should hold: in the full feature set the geometry depends on phase far above chance but is rank-1-dominant, so no stable holonomy forms and RG declines to invent one; in the pre-registered two-dimensional (B_y, B_z) restriction the coupling frame rotates 47° through the year and *replicates across three independent years* ($r = 0.56$, permutation $p = 0.001$), consistent with Russell–McPherron. The net winding is near zero—a reproducible rotation-and-return rather than a topological monodromy—but the operative claim stands: the geoeffective frame is a real, reproducible function of the seasonal condition, and any analysis that pools solar-wind–auroral coupling across seasons in a single fixed frame is wrong by tens of degrees. The seismic and OMNI results are not failures followed by a rescue; they are three coordinates—absent structure, present but rank-1, and present and two-dimensional—that locate precisely where the RG-specific necessity bites, and show that where the preconditions are met, it bites reproducibly.

Non-claims

Non-claim 1. No space-weather forecast, no new heliophysics, no claim about solar-cycle or seasonal mechanisms beyond what is stated as motivation. OMNI is a neutral, dense, coupled, cyclic real dataset used to test the RG necessity phenomena.

Non-claim 2. Pillar 1’s coupling result is conditional-Gaussian inference made one facet of the RG object; its value here is that it is real, lag-validated, and cross-validated, not that it is novel.

Non-claim 3. The absence of a clean holonomy is specific to this feature set and epoch; it is evidence about *preconditions*, not a claim that the solar-wind–magnetosphere geometry is globally flat.

References

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